

Q1	
(a)	Pressure p at depth h in the fluid = $\frac{\text{force } F}{\text{area of base of container } A}$ = $\frac{\text{weight of fluid column } W}{\text{area } A}$ = $\frac{\text{mass of fluid column } m \times \text{acceleration of free fall } g}{A}$ = $\frac{(\text{density } \rho \times \text{volume } V)g}{A}$ = $\frac{\rho Ahg}{A}$ = ρgh
(b)(i)	Let the volume of the object be V. Resultant force $F = \text{weight of object } mg - \text{upthrust } U$ $F = mg - U$ = $(Vd)g - V\rho g$ = $Vdg\left(1 - \frac{\rho}{d}\right)$ = $mg\left(1 - \frac{\rho}{d}\right)$
(b)(ii)	At equilibrium, taking moments about the pivot, Clockwise moment due to sample = anticlockwise moment due to standard mass (Resultant force on sample) $\times$ = (Resultant force on standard mass) $\times$ $mg\left(1 - \frac{\rho}{d}\right) = m_s g\left(1 - \frac{\rho}{d_s}\right)$ $m = m_s\left(1 - \frac{\rho}{d_s}\right) / \left(1 - \frac{\rho}{d}\right)$ = $(0.17851)\left(1 - \frac{1.29}{8493}\right) / \left(1 - \frac{1.29}{940.0}\right)$ = 0.17873 kg (5 d.p.)

